Christopher Moseley - moseleyc@iastate.edu
Lawson Port - lport@iastate.edu

Comprehensive Test Plan for Calculator Application

1. Introduction

The purpose of this test plan is to ensure the overall correctness and functionality of the calculator application, both at the Model code level and the GUI. The testing will confirm that:

1. All mathematical operations behave as expected.
2. The user interface updates correctly in response to user interactions.
3. The system meets the defined use-case requirements.

This test plan delineates the testing process, tools, and methodologies used for validating the application.

2. Scope of Testing

The calculator application will be tested for:

1. **Model**:
    - All mathematical operations are implemented in the CalculatorModel
    - Edge cases (e.g., division by zero, negative inputs).
2. **GUI**:
    - User interaction behavior, including input button responses and display updates.
    - Responses to pressing the operator and function buttons.
    - Styling updates when the operation buttons are pressed.

The Model will be tested using JUNIT tests, and the GUI will be tested using a UI testing tool to simulate user interaction.

3. Testing Approach

The testing will focus on the following areas:

1. **Unit Testing**:
    - They will be tested using JUnit
    - Operations like addition, subtraction, multiplication, division, square, square root, and memory functions will be tested with expected and edge cases.
2. **GUI Testing**:

- The GUI will be tested using the UI automation tool Java Swing Utilities.
- Tests will simulate button clicks and validate the display results.

4. Tools

- **JUnit**
- **Java Swing Utilities**
- **Screenshots**: To verify and document the test outputs.

5. Test Requirements and Test Cases

Test Requirement 1: Core GUI Functionality

These tests verify correct operation and GUI interaction for all basic and advanced calculator functions.

Test Case	Description	Expected Display	Result
TC1	Perform addition: 2 + 3	2 → 3 → 5	Pass
TC2	Perform subtraction: 5 − 2	5 → 2 → 3	Pass
TC3	Perform multiplication: 6 × 7	6 → 7 → 42	Pass
TC4	Perform division: 8 ÷ 2	8 → 2 → 4	Pass
TC5	Square operation: 7^2	7 → 49	Pass
TC6	Square root: $\sqrt{9}$	9 → 3	Pass

TC7	Division by zero: 8 ÷ 0	8 → 0 → Error	Pass
TC16	Invalid sqrt of negative number	-4 → √ → Error	Pass
TC17	Negate number toggle	5 → -5 → 5	Pass
TC18	Memory recall after M+	(3 + 3 = 6 → M+) → MR → shows 0	Pass

---

Test Requirement 2: Only operands and results are displayed

Tests confirm that operator symbols (+, −, ×, ÷) are never shown on screen — only operands and final results appear during interaction.

Test Case	Description	Expected Display Sequence	Result
TC8	Perform addition: 2 + 3	2 → 3 → 5	Pass
TC9	Perform multiplication: 6 × 7	6 → 7 → 42	Pass
TC10	Perform subtraction: 5 − 2	5 → 2 → 3	Pass
TC11	Perform division: 8 ÷ 2	8 → 2 → 4	Pass

Test Requirement 3: Operation button changes state

These tests validate that the operation buttons visually indicate their "pressed" state (e.g., gray background), and return to normal after completion.

Test Case	Description	Expected Behavior	Result
TC12	Press + and complete operation	Turns gray → stays during input → reverts after =	Pass
TC13	Press − and complete operation	Turns gray → stays during input → reverts after =	Pass
TC14	Press × and complete operation	Turns gray → stays during input → reverts after =	Pass
TC15	Press ÷ and complete operation	Turns gray → stays during input → reverts after =	Pass

7. Conclusion

The successfully validated the following: `CalculatorUITest`

1. **Mathematical Operations**: All math-related functionality, including addition, subtraction, multiplication, division, square, and square root, was accurate. Edge cases like division by zero were handled correctly.
2. **Display Behavior**: The display updated correctly during calculations and showed only operands and results, meeting the requirements. No operator symbols (e.g., `+`, `-`) were displayed.
3. **Button State Changes**: Operator buttons (`+`, `-`, `*`,) showed correct visual feedback when pressed and reverted to normal upon completion of a calculation. `/`

No issues or defects were identified. The GUI behaves as expected under all tested scenarios.